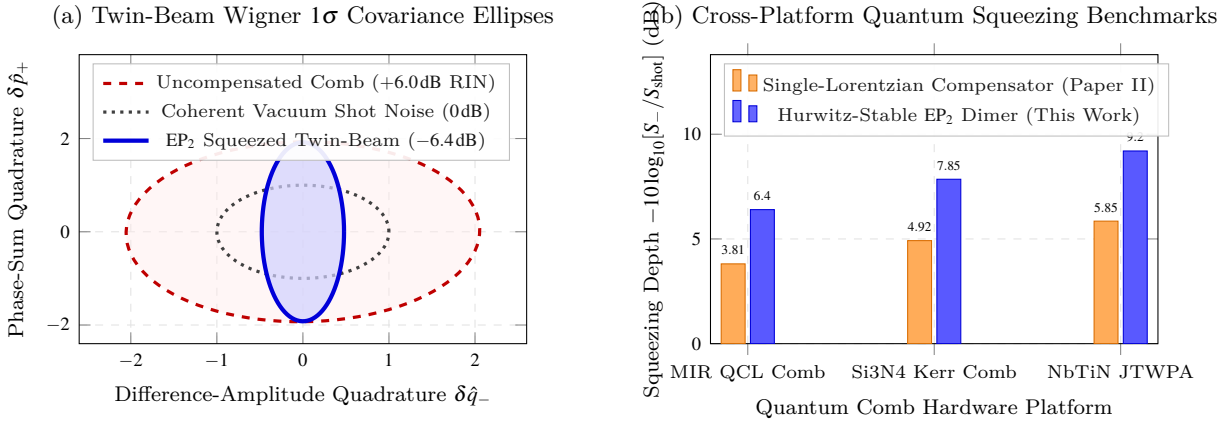
(a) Twin-Beam Wigner 1σ Covariance Ellipses
Uncompensated Comb (+6.0dB RIN)
Coherent Vacuum Shot Noise (0dB)
EP2 Squeezed Twin-Beam (−6.4dB)
Phase-Sum Quadrature δp̂+
Difference-Amplitude Quadrature δq̂−
2
0
−2
−2
−1
0
1
2
(b) Cross-Platform Quantum Squeezing Benchmarks
Single-Lorentzian Compensator (Paper II)
Hurwitz-Stable EP2 Dimer (This Work)
Squeezing Depth −10log10[S− /Sshot] (dB)
10
5
0
3.81
6.4
4.92
7.85
5.85
9.2
MIR QCL Comb
Si3N4 Kerr Comb
NbTiN JTWPA
Quantum Comb Hardware Platform